

Literature Status: Strongly Norm Attaining Lipschitz Maps

Automatic Math Discovery Run `fa_banach_001`

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Status

This is a **literature_already_answered** packet. It records that the main open-problem signals extracted from arXiv:1807.03363 were explicitly answered or sharply delimited by the later paper arXiv:1907.07698. It is not a new proof packet.

Source Paper and Questions

The source paper is:

B. Cascales, R. Chiclana, L. C. Garcia-Lirola, M. Martin, and A. Rueda Zoca, *On strongly norm attaining Lipschitz maps*, arXiv:1807.03363.

In the introduction, the authors note that if $\text{SA}(M, Y)$ is dense in $\text{Lip}(M, Y)$ for every Banach space Y , then $\mathcal{F}(M)$ has Lindenstrauss property A. They state that the converse is not known and that, as far as they know, the Radon–Nikodym property of $\mathcal{F}(M)$ could be necessary for this density. This is the source-paper signal covered by the present literature-status packet.

Supporting Answer Paper

The supporting paper is:

R. Chiclana, L. C. Garcia-Lirola, M. Martin, A. Rueda Zoca, and P. Tradacete, *Examples and applications of the density of strongly norm attaining Lipschitz maps*, arXiv:1907.07698.

The introduction of arXiv:1907.07698 formulates the relevant issues as Questions Q1–Q4. In particular, Q1 asks whether $\mathcal{F}(M)$ must have the RNP when $\text{SA}(M, Y)$ is dense in $\text{Lip}(M, Y)$ for every Banach space Y ; Q2 asks whether

$$B_{\mathcal{F}(M)} = \overline{\text{co}}(\text{str-exp}(B_{\mathcal{F}(M)}))$$

or Gromov concavity suffices for density of scalar strongly norm attaining Lipschitz maps; Q3 asks for the converse implication; and Q4 asks whether the presence of an isomorphic, or even isometric, $L_1[0, 1]$ copy forces failure of density.

Answer Identification

Theorem `theo:lluvia` of arXiv:1907.07698 constructs metric spaces M_p , $p = 1, 2$, such that $\text{SA}(M_p, Y)$ is dense in $\text{Lip}(M_p, Y)$ for every Banach space Y , while each M_p contains an isometric copy of $[0, 1]$. Consequently $\mathcal{F}(M_p)$ contains a copy of $L_1[0, 1]$ and fails the RNP. This gives a negative answer to the source paper’s RNP-necessity question, and also a negative answer to Q4.

Theorem `theo:toro` of arXiv:1907.07698 treats the Euclidean unit circle $\mathbb{T}$. It proves that $\text{SA}(\mathbb{T}, \mathbb{R})$ is not dense in $\text{Lip}(\mathbb{T}, \mathbb{R})$, although every molecule is a strongly exposed point of $B_{\mathcal{F}(\mathbb{T})}$. Thus $B_{\mathcal{F}(\mathbb{T})}$ is the closed convex hull of its strongly exposed points, and $\mathbb{T}$ is Gromov concave. This gives a negative answer to Q2.

For Q3, Theorem `theorem:clcvstrepcompact` of arXiv:1907.07698 proves that if M is compact and $\text{SA}(M, \mathbb{R})$ is dense in $\text{Lip}(M, \mathbb{R})$, then

$$B_{\mathcal{F}(M)} = \overline{\text{co}}(\text{str-exp}(B_{\mathcal{F}(M)})).$$

Corollary `coro:clcvstrepbouncompact` extends this conclusion to boundedly compact metric spaces.

Scope Limitations

This packet clears the RNP/density branch from arXiv:1807.03363 and records closely related consequences developed in arXiv:1907.07698. It does not claim to solve the newer residual questions raised inside arXiv:1907.07698, such as the bi-Lipschitz closed-subset variants of the RNP question or the question whether an equivalent metric on $[0, 1]$ can yield density of $\text{SA}([0, 1], d, \mathbb{R})$. It also does not record a later answer to the separate octahedrality question mentioned in arXiv:1807.03363.

Search Evidence

Local cheap indexes for arXiv:1807.03363, “strongly norm attaining Lipschitz maps”, “Lipschitz-free”, and “RNP” had no prior exact packet at reservation time. Direct source inspection found arXiv:1907.07698 as a later paper by overlapping authors that explicitly cites arXiv:1807.03363 and formulates the relevant density/RNP question as Q1 before answering it as described above.

References

- [1] B. Cascales, R. Chiclana, L. C. Garcia-Lirola, M. Martin, and A. Rueda Zoca, *On strongly norm attaining Lipschitz maps*, arXiv:1807.03363.
- [2] R. Chiclana, L. C. Garcia-Lirola, M. Martin, A. Rueda Zoca, and P. Tradacete, *Examples and applications of the density of strongly norm attaining Lipschitz maps*, arXiv:1907.07698.